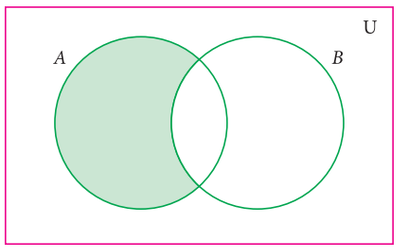
### 1.5.4 Difference of Two Sets

Let $A$ and $B$ be two sets, the difference of sets $A$ and $B$ is the set of all elements which are in $A$, but not in $B$. It is denoted by $A-B$ or A\B and read as $A$ difference $B$.

In symbol, $A-B = \{ x : x \in A \text{ and } x \notin B\}$

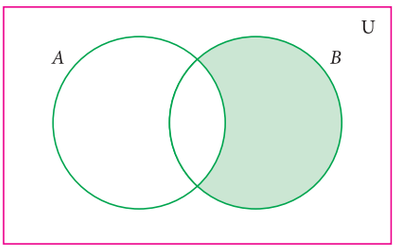
$B-A = \{ y : y \in B \text{ and } y \notin A\}$.

**Venn diagram for set difference**



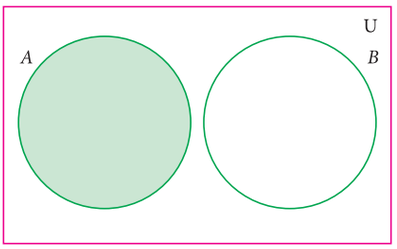

$A-B$

Fig. 1.13




$B-A$

Fig. 1.14




$A-B$

Fig. 1.15

**Example 1.14**

If $A=\{-3, -2, 1, 4\}$ and $B=\{0, 1, 2, 4\}$, find (i) $A-B$ (ii) $B-A$.

***Solution***

$A-B = \{-3, -2, 1, 4\} - \{0, 1, 2, 4\} = \{-3, -2\}$

$B-A = \{0, 1, 2, 4\} - \{-3, -2, 1, 4\} = \{0, 2\}$

**Note**

- $A' = U - A$
- $A-B = A \cap B'$
- $A-A = \varnothing$
- $A-\varnothing = A$
- $A-B = B-A \Leftrightarrow A=B$
- $A-B = A$ and $B-A = B$ if $A \cap B = \varnothing$

### 1.5.5 Symmetric Difference of Sets

The symmetric difference of two sets $A$ and $B$ is the set $(A-B)\cup(B-A)$. It is denoted by $A \Delta B$.

$A \Delta B = \{ x : x \in A-B \text{ or } x \in B-A\}$

**Example 1.15**

If $A = \{6, 7, 8, 9\}$ and $B=\{8, 10, 12\}$, find $A \Delta B$.

***Solution***

$A-B = \{6, 7, 9\}$

$B-A = \{10, 12\}$

$A \Delta B = (A-B)\cup(B-A) = \{6, 7, 9\}\cup\{10,12\}$

$A \Delta B = \{6, 7, 9, 10, 12\}$.

**Thinking Corner**

What is $(A - B) \cap (B - A)$ ?



1-SET LANGUAGE.indd 16 02/12/2022 21:30:52